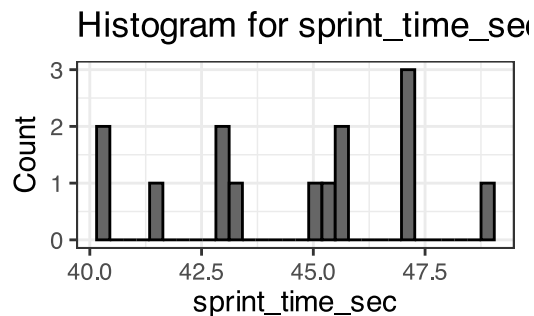
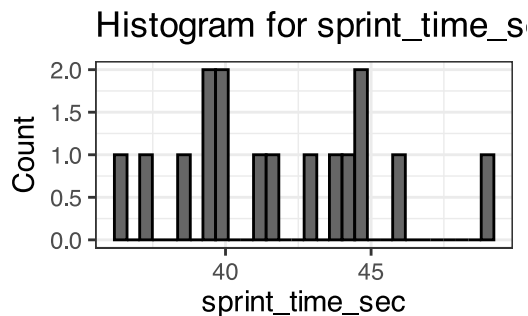
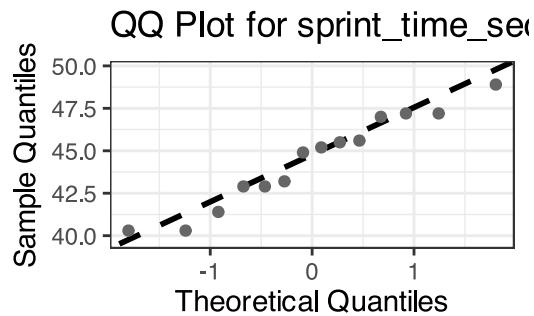
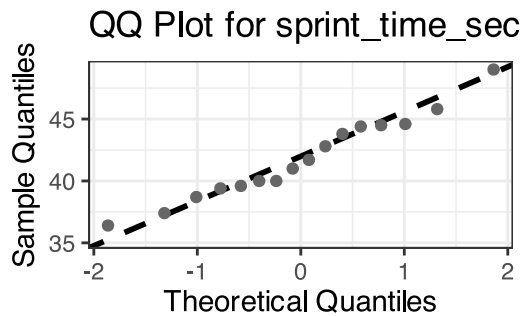


Example 1

- Every cadet runs a timed sprint course as part of camp trials.
- The coaches want us to determine if the Thunder Squadron and Lightning Squadron are finishing the sprint course at different times.
- Checking normality,

```
wonderbolt_camp |> independent_qq(outcome = sprint_time_sec,  
                                   grouping = squadron)
```



- Using the `independent_medians()` function,

```
wonderbolt_camp |> independent_medians(outcome = sprint_time_sec,  
                                       grouping = squadron,  
                                       alternative = "two",  
                                       m = 0,  
                                       alpha = 0.05)
```

Independent medians

M[Lightning Squadron] = 45.05

M[Thunder Squadron] = 41.35

Hodges-Lehmann estimate of median difference: 2.8

95% CI for M[Lightning Squadron] - M[Thunder Squadron]: (0.4, 5.5)

Hypotheses:

$H_0: M[\text{Lightning Squadron}] - M[\text{Thunder Squadron}] = 0$

$H_1: M[\text{Lightning Squadron}] - M[\text{Thunder Squadron}] \neq 0$

Test statistic: $W = 168$

p-value: 0.019

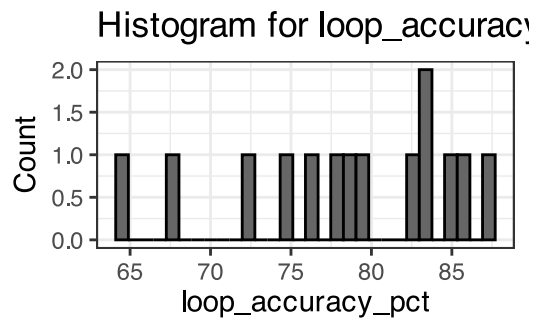
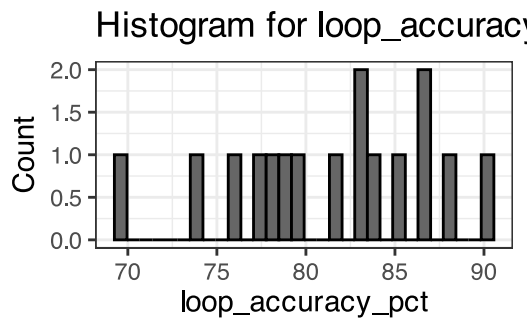
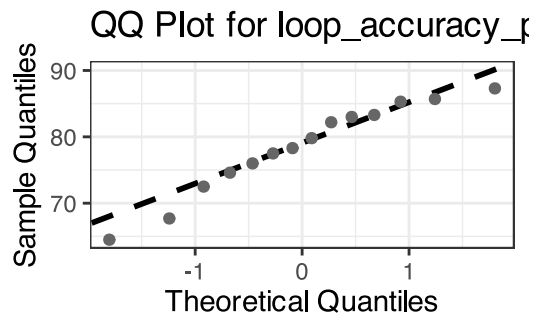
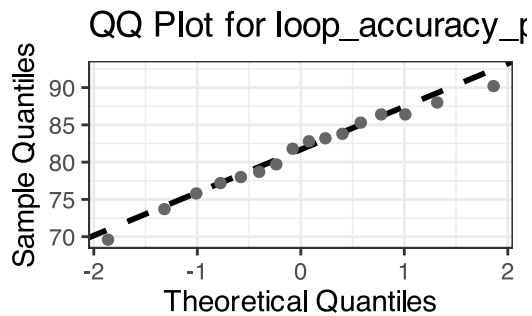
Conclusion: Reject the null hypothesis ($p = 0.019 < \alpha = 0.050$)

- **Hypotheses:**
- **Test Statistic and p -Value**
- **Rejection Region**
- **Conclusion and interpretation**

Example 2

- At camp, judges score each cadet's aerial loop maneuver on a 0–100 accuracy scale during trials.
- Based on her gut feeling, the head judge believes that, on average, the Thunder Squadron is performing more accurate loop maneuvers.
- Checking normality,

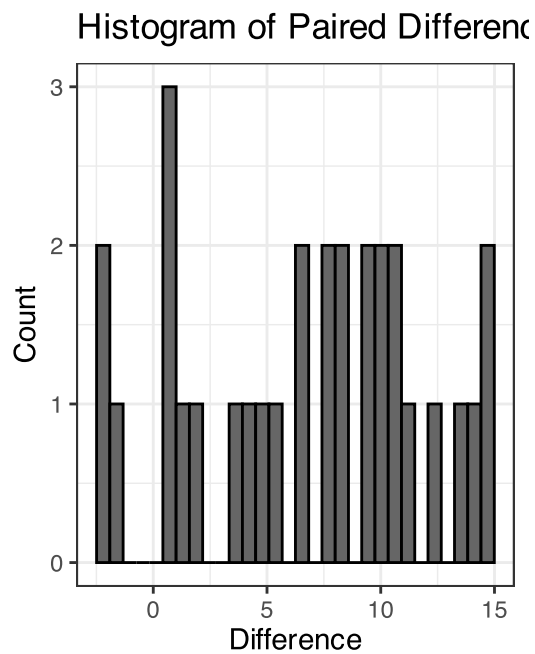
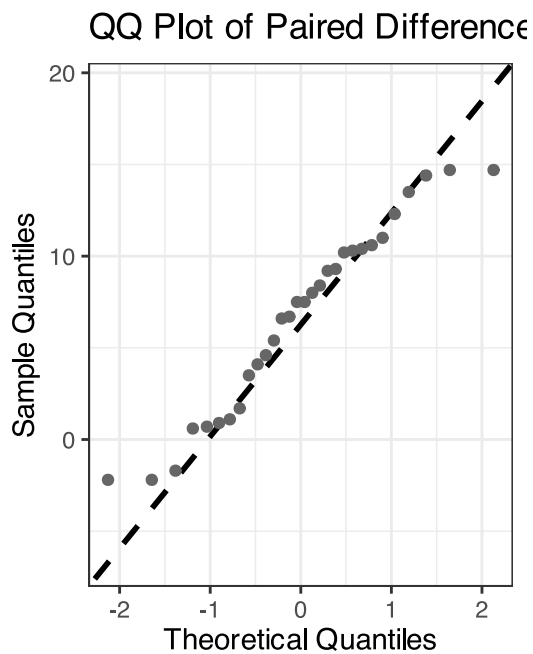
```
wonderbolt_camp |> independent_qq(outcome = loop_accuracy_pct,  
                                grouping = squadron)
```



Example 3

- Each cadet's aerial accuracy is scored when they arrive at camp, and scored again at the end after two weeks of drills.
- Coaches want to know whether cadets are actually flying more accurately by the end of camp.
- Checking normality,

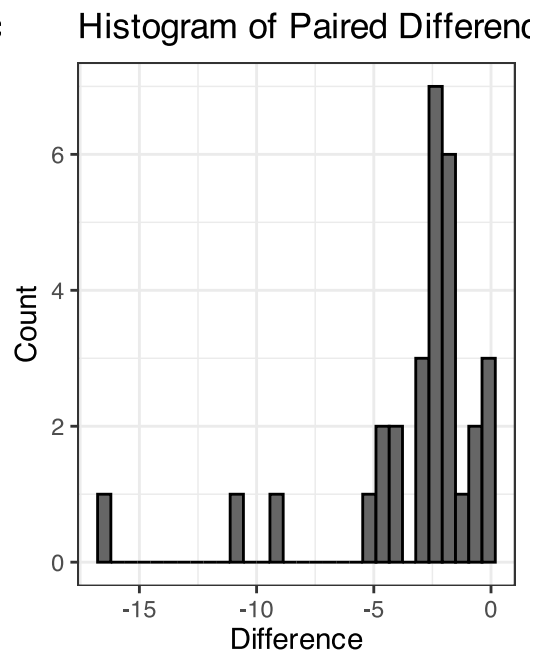
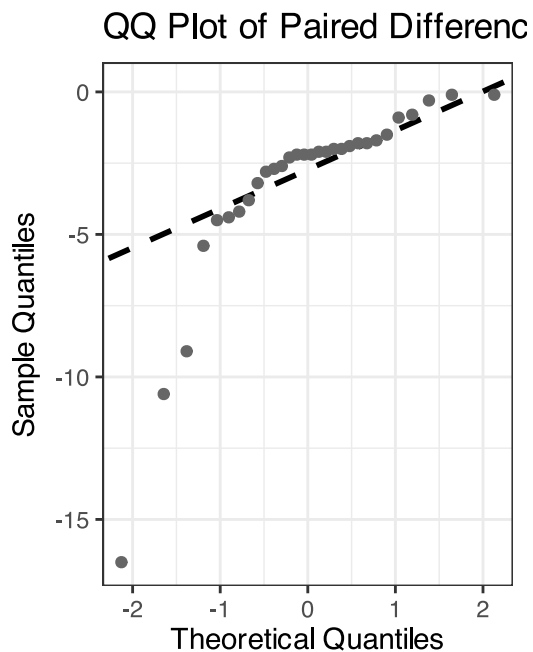
```
wonderbolt_camp |> dependent_qq(col1 = accuracy_score_post,  
                                col2 = accuracy_score_pre)
```



Example 4

- Each cadet's recovery time after a hard trial is timed at the start of camp and again at the end.
- Coaches want to know whether cadets are recovering faster by the end of camp.
- Checking normality,

```
wonderbolt_camp |> dependent_qq(coll = recovery_time_min_post,  
                                col2 = recovery_time_min_pre)
```



- Using the `dependent_medians()` function,

```
wonderbolt_camp |> dependent_medians(coll = recovery_time_min_post,  
                                     col2 = recovery_time_min_pre,  
                                     alternative = "less",  
                                     m = 0,  
                                     alpha = 0.05)
```

Paired (dependent) medians

Complete pairs: 30

Effective pairs (non-zero differences): 30

$M[\text{recovery_time_min_post}] = 7.6$

$M[\text{recovery_time_min_pre}] = 12.6$

Hodges-Lehmann estimate of median difference: -2.4

95% CI for $M[\text{recovery_time_min_post}] - M[\text{recovery_time_min_pre}]$: (-3.3, -1.95)

Hypotheses:

$H_0: M[\text{recovery_time_min_post}] - M[\text{recovery_time_min_pre}] \geq 0$

$H_1: M[\text{recovery_time_min_post}] - M[\text{recovery_time_min_pre}] < 0$

Test statistic: $T = 0$

p-value: < 0.001

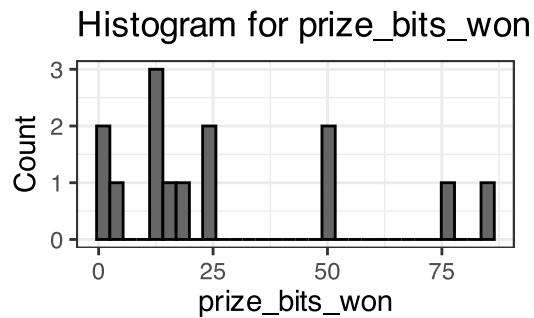
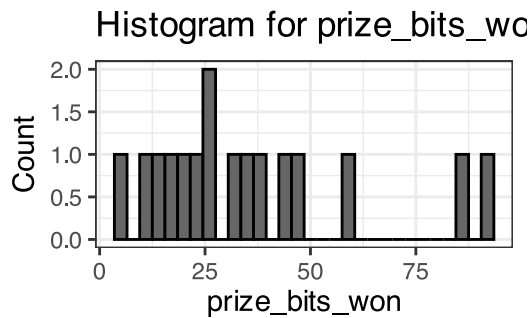
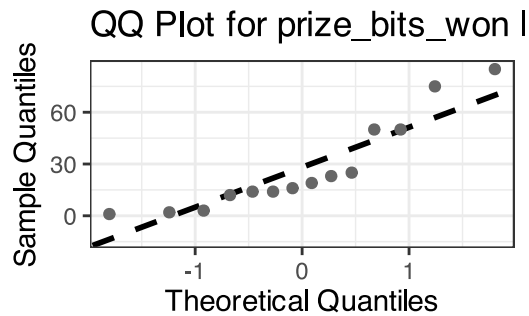
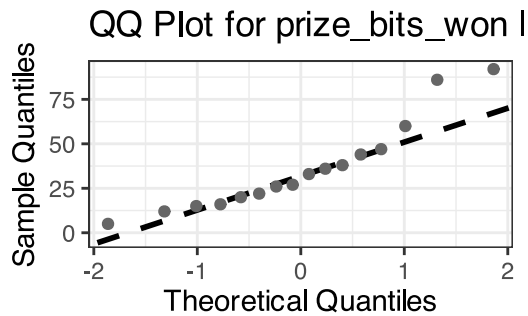
Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- **Hypotheses:**
- **Test Statistic and p -Value**
- **Rejection Region**
- **Conclusion and interpretation**

Let's Practice! 1

- Spectators toss bits to cadets who pull off a standout run during trials.
- Camp organizers want to know if the squadrons are earning prize bits equally.
- Checking normality,

```
wonderbolt_camp |> independent_qq(outcome = prize_bits_won,
                                   grouping = squadron)
```



- Analyzing using the *t*-test,

```
wonderbolt_camp |> independent_means(outcome = prize_bits_won,
                                      grouping = squadron,
                                      alternative = "two",
                                      mu = 0,
                                      alpha = 0.05)
```

Independent means (Welch's t-test)

$\bar{x}$ [Lightning Squadron] = 27.7857

$\bar{x}$ [Thunder Squadron] = 36.1875

Point estimate: $\bar{x}$ [Lightning Squadron] - $\bar{x}$ [Thunder Squadron] = -8.4018

95% CI for μ [Lightning Squadron] - μ [Thunder Squadron]: (-27.9469, 11.1433)

Hypotheses:

$H_0: \mu[\text{Lightning Squadron}] - \mu[\text{Thunder Squadron}] = 0$

$H_1: \mu[\text{Lightning Squadron}] - \mu[\text{Thunder Squadron}] \neq 0$

Test statistic: $t(26.86) = -0.882$

p-value: 0.385

Conclusion: Fail to reject the null hypothesis ($p = 0.385 \geq \alpha = 0.050$)

- Analyzing using the Wilcoxon rank sum,

```
wonderbolt_camp |> independent_medians(outcome = prize_bits_won,  
                                         grouping = squadron,  
                                         alternative = "two",  
                                         m = 0,  
                                         alpha = 0.05)
```

Independent medians

$M[\text{Lightning Squadron}] = 17.5$

$M[\text{Thunder Squadron}] = 30$

Hodges-Lehmann estimate of median difference: -10

95% CI for $M[\text{Lightning Squadron}] - M[\text{Thunder Squadron}]$: (-24, 6)

Hypotheses:

$H_0: M[\text{Lightning Squadron}] - M[\text{Thunder Squadron}] = 0$

$H_1: M[\text{Lightning Squadron}] - M[\text{Thunder Squadron}] \neq 0$

Test statistic: $W = 79$

p-value: 0.176

Conclusion: Fail to reject the null hypothesis ($p = 0.176 \geq \alpha = 0.050$)

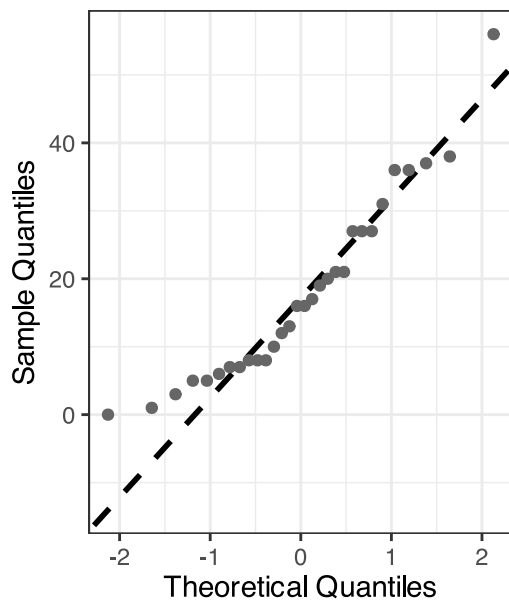
- **Hypotheses:**
- **Test Statistic and p -Value**
- **Rejection Region**
- **Conclusion and interpretation**

Let's Practice! 2

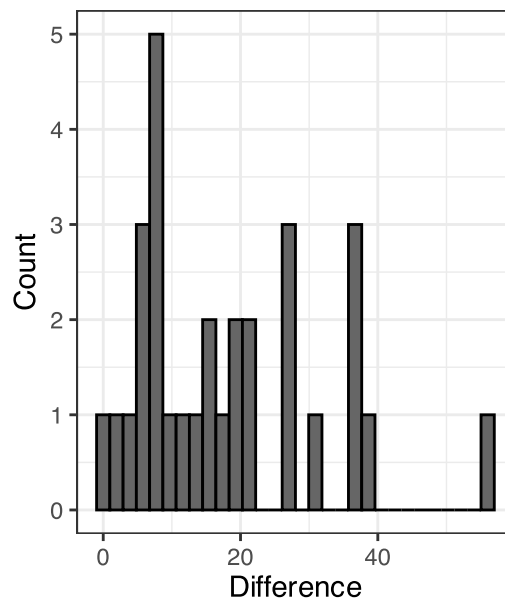
- Spectators tip cadets with bits after their runs. The amount tipped is recorded once early in camp and once again near the end.
- Camp organizers want to know whether cadets are earning more in tips as they improve over camp.
- Checking normality,

```
wonderbolt_camp |> dependent_qq(coll = tip_bits_post,  
                                col2 = tip_bits_pre)
```

QQ Plot of Paired Difference



Histogram of Paired Difference



- Analyzing using the paired *t*-test,

```
wonderbolt_camp |> dependent_means(coll = tip_bits_post,  
                                   col2 = tip_bits_pre,  
                                   alternative = "greater",  
                                   mu = 0,  
                                   alpha = 0.05)
```

Paired (dependent) means

Complete pairs: 30

Point estimate: $\bar{x}[\text{tip_bits_post}] - \bar{x}[\text{tip_bits_pre}] = 17.9333$

SD of differences: 13.4982

95% CI for $\mu[\text{tip_bits_post}] - \mu[\text{tip_bits_pre}]$: (12.893, 22.9737)

Hypotheses:

$H_0: \mu[\text{tip_bits_post}] - \mu[\text{tip_bits_pre}] \leq 0$

$H_1: \mu[\text{tip_bits_post}] - \mu[\text{tip_bits_pre}] > 0$

Test statistic: $t(29) = 7.277$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- Analyzing using the Wilcoxon signed rank,

```
wonderbolt_camp |> dependent_medians(coll = tip_bits_post,
                                     col2 = tip_bits_pre,
                                     alternative = "greater",
                                     m = 0,
                                     alpha = 0.05)
```

Paired (dependent) medians

Complete pairs: 30

Effective pairs (non-zero differences): 29

$M[\text{tip_bits_post}] = 23.5$

$M[\text{tip_bits_pre}] = 9$

Hodges-Lehmann estimate of median difference: 17

95% CI for $M[\text{tip_bits_post}] - M[\text{tip_bits_pre}]$: (12, 22)

Hypotheses:

$H_0: M[\text{tip_bits_post}] - M[\text{tip_bits_pre}] \leq 0$

$H_1: M[\text{tip_bits_post}] - M[\text{tip_bits_pre}] > 0$

Test statistic: $T = 464$

p-value: < 0.001

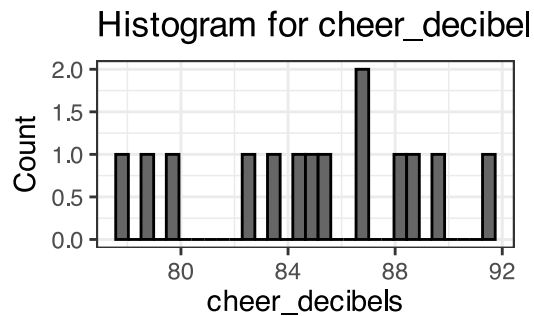
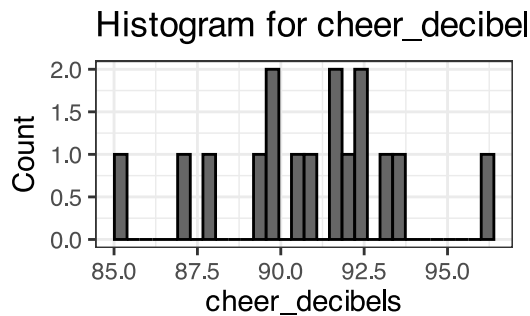
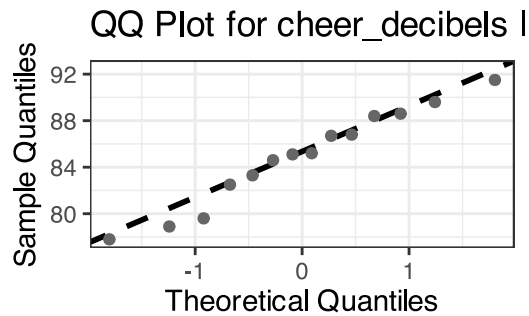
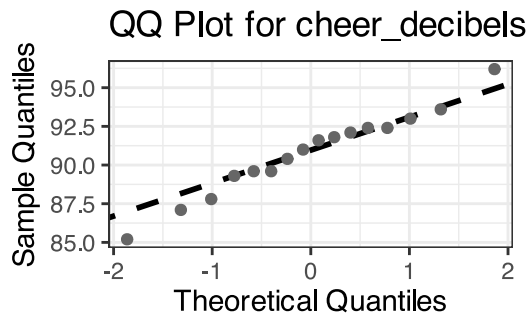
Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- **Hypotheses:**
- **Test Statistic and p -Value**
- **Rejection Region**
- **Conclusion and interpretation**

Let's Practice! 3

- A sound crew records the peak crowd noise (in decibels) during each cadet's run.
- Camp organizers believe that the Thunder Squadron is inspiring a higher level of noise.
- Checking normality,

```
wonderbolt_camp |> independent_qq(outcome = cheer_decibels,  
                                   grouping = squadron)
```



- Using the independent_means() function,

```
wonderbolt_camp |> independent_means(outcome = cheer_decibels,  
                                     grouping = squadron,  
                                     alternative = "less",  
                                     mu = 0,  
                                     alpha = 0.05)
```

Independent means (Welch's t-test)

$\bar{x}$ [Lightning Squadron] = 84.9

$\bar{x}$ [Thunder Squadron] = 90.8187

Point estimate: $\bar{x}$ [Lightning Squadron] - $\bar{x}$ [Thunder Squadron] = -5.9187

95% CI for μ [Lightning Squadron] - μ [Thunder Squadron]: (-8.6025, -3.235)

Hypotheses:

$H_0: \mu[\text{Lightning Squadron}] - \mu[\text{Thunder Squadron}] \geq 0$

$H_1: \mu[\text{Lightning Squadron}] - \mu[\text{Thunder Squadron}] < 0$

Test statistic: $t(21.92) = -4.575$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- Using the `independent_medians()` function,

```
wonderbolt_camp |> independent_medians(outcome = cheer_decibels,  
                                         grouping = squadron,  
                                         alternative = "less",  
                                         m = 0,  
                                         alpha = 0.05)
```

Independent medians

$M[\text{Lightning Squadron}] = 85.15$

$M[\text{Thunder Squadron}] = 91.3$

Hodges-Lehmann estimate of median difference: -5.65

95% CI for $M[\text{Lightning Squadron}] - M[\text{Thunder Squadron}]$: (-8.5, -2.9)

Hypotheses:

$H_0: M[\text{Lightning Squadron}] - M[\text{Thunder Squadron}] \geq 0$

$H_1: M[\text{Lightning Squadron}] - M[\text{Thunder Squadron}] < 0$

Test statistic: $W = 21.5$

p-value: < 0.001

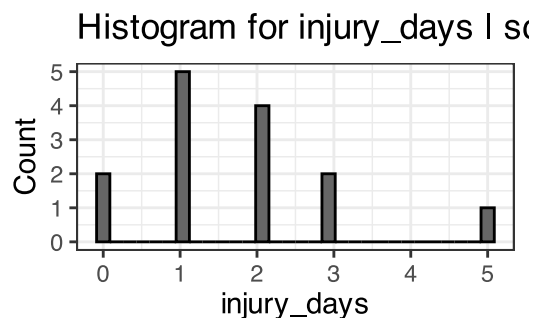
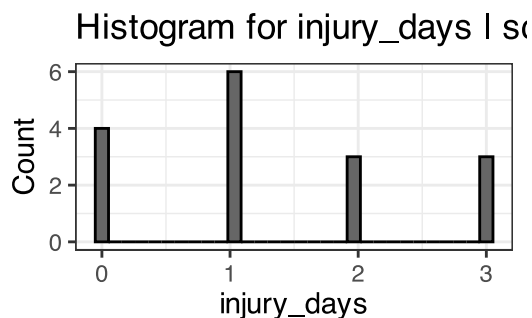
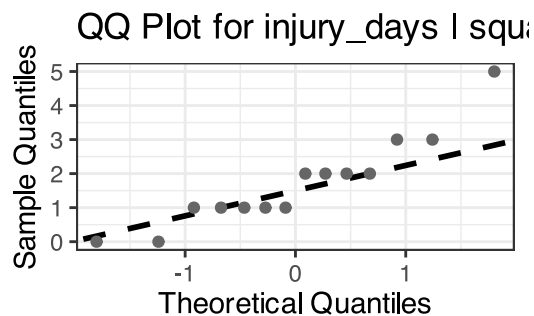
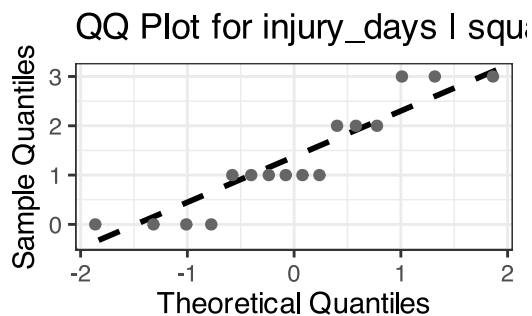
Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- **Hypotheses:**
- **Test Statistic and p -Value**
- **Rejection Region**
- **Conclusion and interpretation**

Let's Practice! 4

- The camp medic tracks how many days each cadet spends recovering from an injury over the two-week camp.
- Coaches want to know if the two training styles lead to the same number of injury days.
- Checking normality,

```
wonderbolt_camp |> independent_qq(outcome = injury_days,
                                   grouping = squadron)
```



- Using the `independent_means()` function,

```
wonderbolt_camp |> independent_means(outcome = injury_days,
                                     grouping = squadron,
                                     alternative = "two",
                                     mu = 0,
                                     alpha = 0.05)
```

Independent means (Welch's t-test)

$\bar{x}$ [Lightning Squadron] = 1.7143

$\bar{x}$ [Thunder Squadron] = 1.3125

Point estimate: $\bar{x}$ [Lightning Squadron] - $\bar{x}$ [Thunder Squadron] = 0.4018

95% CI for μ [Lightning Squadron] - μ [Thunder Squadron]: (-0.515, 1.3186)

Hypotheses:

$H_0: \mu[\text{Lightning Squadron}] - \mu[\text{Thunder Squadron}] = 0$

$H_1: \mu[\text{Lightning Squadron}] - \mu[\text{Thunder Squadron}] \neq 0$

Test statistic: $t(25.11) = 0.902$

p-value: 0.375

Conclusion: Fail to reject the null hypothesis ($p = 0.375 \geq \alpha = 0.050$)

- Using the `independent_medians()` function,

```
wonderbolt_camp |> independent_medians(outcome = injury_days,  
                                         grouping = squadron,  
                                         alternative = "two",  
                                         m = 0,  
                                         alpha = 0.05)
```

Independent medians

$M[\text{Lightning Squadron}] = 1.5$

$M[\text{Thunder Squadron}] = 1$

Hodges-Lehmann estimate of median difference: 0

95% CI for $M[\text{Lightning Squadron}] - M[\text{Thunder Squadron}]$: (-1, 1)

Hypotheses:

$H_0: M[\text{Lightning Squadron}] - M[\text{Thunder Squadron}] = 0$

$H_1: M[\text{Lightning Squadron}] - M[\text{Thunder Squadron}] \neq 0$

Test statistic: $W = 130$

p-value: 0.451

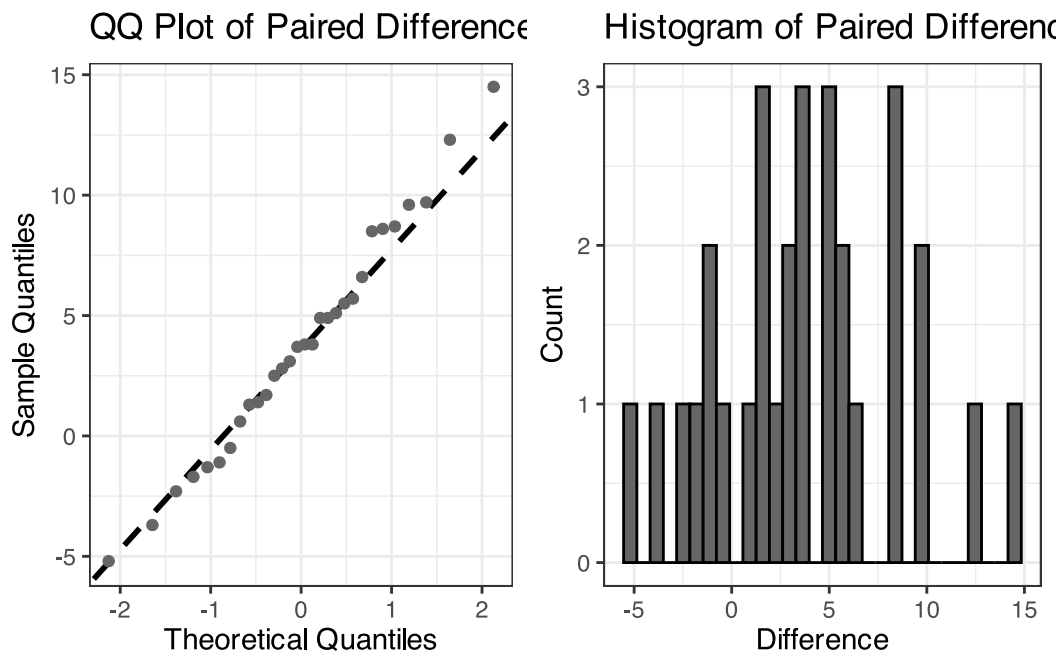
Conclusion: Fail to reject the null hypothesis ($p = 0.451 \geq \alpha = 0.050$)

- **Hypotheses:**
- **Test Statistic and p -Value**
- **Rejection Region**
- **Conclusion and interpretation**

Let's Practice! 5

- Each cadet's wing-flap rate while hovering in place is measured on arrival, then measured again after two weeks of drills.
- Coaches want to know whether training actually changes a cadet's wing-flap rate.
- Checking normality,

```
wonderbolt_camp |> dependent_qq(col1 = wing_flap_rate_post,  
                                col2 = wing_flap_rate_pre)
```



- Using the `dependent_means()` function,

```
wonderbolt_camp |> dependent_means(col1 = wing_flap_rate_post,  
                                   col2 = wing_flap_rate_pre,  
                                   alternative = "greater",  
                                   mu = 0,  
                                   alpha = 0.05)
```

Paired (dependent) means

Complete pairs: 30

Point estimate: $\bar{x}[\text{wing_flap_rate_post}] - \bar{x}[\text{wing_flap_rate_pre}] = 3.7833$

SD of differences: 4.7192

95% CI for $\mu[\text{wing_flap_rate_post}] - \mu[\text{wing_flap_rate_pre}]$: (2.0212, 5.5455)

Hypotheses:

$H_0: \mu[\text{wing_flap_rate_post}] - \mu[\text{wing_flap_rate_pre}] \leq 0$

$H_1: \mu[\text{wing_flap_rate_post}] - \mu[\text{wing_flap_rate_pre}] > 0$

Test statistic: $t(29) = 4.391$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- Using the `dependent_medians()` function,

```
wonderbolt_camp |> dependent_medians(col1 = wing_flap_rate_post,
                                     col2 = wing_flap_rate_pre,
                                     alternative = "greater",
                                     m = 0,
                                     alpha = 0.05)
```

Paired (dependent) medians

Complete pairs: 30

Effective pairs (non-zero differences): 30

$M[\text{wing_flap_rate_post}] = 55.35$

$M[\text{wing_flap_rate_pre}] = 50.65$

Hodges-Lehmann estimate of median difference: 3.7

95% CI for $M[\text{wing_flap_rate_post}] - M[\text{wing_flap_rate_pre}]$: (1.9, 5.5)

Hypotheses:

$H_0: M[\text{wing_flap_rate_post}] - M[\text{wing_flap_rate_pre}] \leq 0$

$H_1: M[\text{wing_flap_rate_post}] - M[\text{wing_flap_rate_pre}] > 0$

Test statistic: $T = 405.5$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

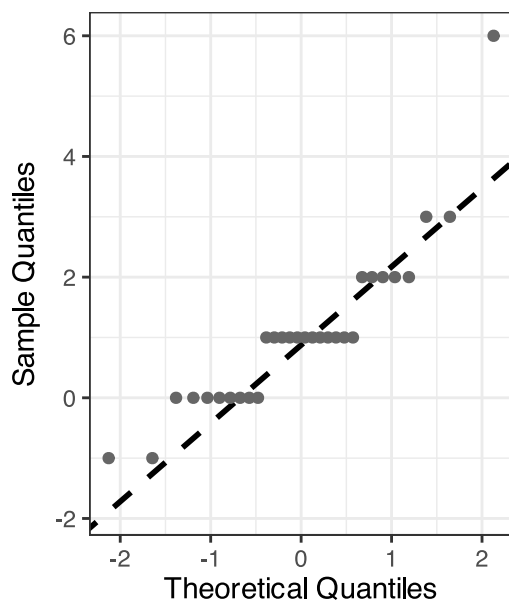
- **Hypotheses:**
- **Test Statistic and p -Value**
- **Rejection Region**
- **Conclusion and interpretation**

Let's Practice! 6

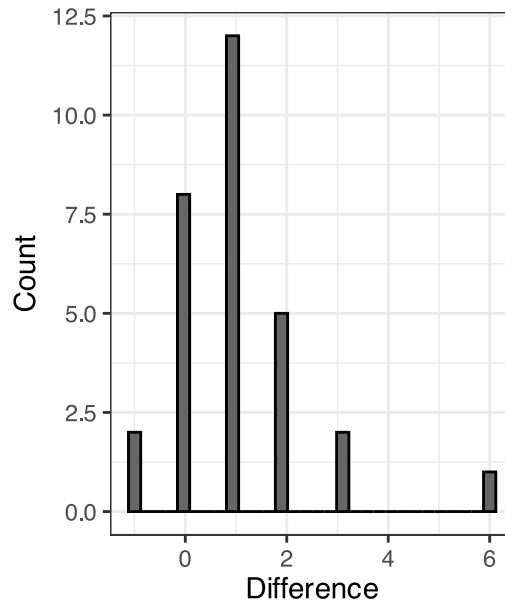
- Each cadet rates how confident they feel about their flying on a self-reported scale, once at the start of camp and once at the end.
- Camp staff want to know whether two weeks of training changes how confident cadets feel.
- Checking normality,

```
wonderbolt_camp |> dependent_qq(col1 = confidence_rating_post,  
                                col2 = confidence_rating_pre)
```

QQ Plot of Paired Difference



Histogram of Paired Difference



- Using the `dependent_means()` function,

```
wonderbolt_camp |> dependent_means(col1 = confidence_rating_post,  
                                   col2 = confidence_rating_pre,  
                                   alternative = "greater",  
                                   mu = 0,  
                                   alpha = 0.05)
```

Paired (dependent) means

Complete pairs: 30

Point estimate: $\bar{x}[\text{confidence_rating_post}] - \bar{x}[\text{confidence_rating_pre}] = 1.0667$

SD of differences: 1.3629

95% CI for $\mu[\text{confidence_rating_post}] - \mu[\text{confidence_rating_pre}]$: (0.5578,

1.5756)

Hypotheses:

$H_0: \mu[\text{confidence_rating_post}] - \mu[\text{confidence_rating_pre}] \leq 0$

$H_1: \mu[\text{confidence_rating_post}] - \mu[\text{confidence_rating_pre}] > 0$

Test statistic: $t(29) = 4.287$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- Using the `dependent_medians()` function,

```
wonderbolt_camp |> dependent_medians(col1 = confidence_rating_post,
                                     col2 = confidence_rating_pre,
                                     alternative = "greater",
                                     m = 0,
                                     alpha = 0.05)
```

Paired (dependent) medians

Complete pairs: 30

Effective pairs (non-zero differences): 22

$M[\text{confidence_rating_post}] = 7$

$M[\text{confidence_rating_pre}] = 6$

Hodges-Lehmann estimate of median difference: 1

95% CI for $M[\text{confidence_rating_post}] - M[\text{confidence_rating_pre}]$: (0.5, 1.5)

Hypotheses:

$H_0: M[\text{confidence_rating_post}] - M[\text{confidence_rating_pre}] \leq 0$

$H_1: M[\text{confidence_rating_post}] - M[\text{confidence_rating_pre}] > 0$

Test statistic: $T = 398$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- **Hypotheses:**
- **Test Statistic and p -Value**
- **Rejection Region**
- **Conclusion and interpretation**